

NATHAN GLADSTONE

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech) August 2025 – May 2028 (Expected)
BS in Mechanical Engineering — GPA: 3.3
Relevant Coursework: Mechanical Design, Mechanical Vibrations, Mechanics of Deformable Bodies, Fluid Dynamics, Thermodynamics, Dynamics, Engineering Analysis Using Numerical Methods

James Madison University (JMU) August 2023 – May 2025
Engineering — GPA: 3.5 — Honors College (completed two years prior to transferring to Virginia Tech)

WORK EXPERIENCE

Mechanical Engineering Contractor May 2026 – Present
AERO LLC *Manassas, VA*

- Developed a hinge assembly for an aircraft in Autodesk Fusion, carrying 4 concepts through detailed CAD and selecting a design that reduced the mechanism from 2 degrees of freedom to 1. Applied DFM throughout, using off-the-shelf fasteners to simplify machining and assembly.
- Designed a sleeve-style wing joint that adds structural support and acts as a fairing to reduce wing vibration, and began 3D printing prototypes of both the hinge and sleeve.
- Validated 2D airfoil computational fluid dynamics (CFD) simulations in STAR-CCM+ by comparing results to published data across 4 cases at different angles of attack, Mach numbers, and Reynolds numbers.
- Automated the CFD workflow with Bash scripts and Java macros on a Linux high-performance computing (HPC) cluster. Setup dropped from 10 minutes to under 2 per run and the tools are now used company-wide across about 30 simulations.

Materials Engineer Intern May 2025 – August 2025
ECS *Manassas, VA*

- Performed 100+ ASTM materials tests on concrete and soil for construction projects that have since been completed, assessing compressive strength, density, and durability.
- Prepared 50+ technical reports of test data used to evaluate structural integrity and compatibility.
- Presented test results and findings to engineers, technicians, and project managers to guide construction decisions.

PROJECTS & LEADERSHIP

RoboGrinder – RoboMaster Design Team at Virginia Tech August 2025 – Present

- Placed 2nd in North America at RoboMaster 2026 with a 50-person multidisciplinary team.
- Designed a custom holonomic chassis with built-in suspension to meet weight and load constraints while keeping the robot stable and controllable on uneven terrain.
- Modeled 20+ sheet metal and extruded parts in SolidWorks for the robot's mechanical systems with DFM in mind.

VEX Robotics – Mentor, Team 2145 April 2023 – Present

- Mentored 5 teams (about 30 students) in the 2145 organization on robot design and match strategy. One team reached the semifinals at the VEX Robotics World Championship.

VEX Robotics – Team 2145Z August 2016 – April 2023

- Designed and built lift, shooter, and transmission systems across 7 competition seasons, fabricating parts with CNC machining and laser cutting.
- Led the team's mechanical design, strategy, and engineering analysis, winning 4 Virginia State Championships, competing at 5 World Championships, and earning the Inspire Award at the VEX Robotics World Championship.

SKILLS

CAD & Design: SolidWorks, Autodesk Fusion (Fusion 360), engineering drawings, sheet metal design, rapid prototyping
Simulation & Analysis: STAR-CCM+, Computational Fluid Dynamics (CFD), MATLAB
Programming & Software: Java, Linux, Bash scripting, Slurm, HPC, Microsoft Office (Word, Excel, PowerPoint)
Manufacturing: Design for Manufacturing (DFM), 3D printing, CNC machining, laser cutting, soldering, hand tools